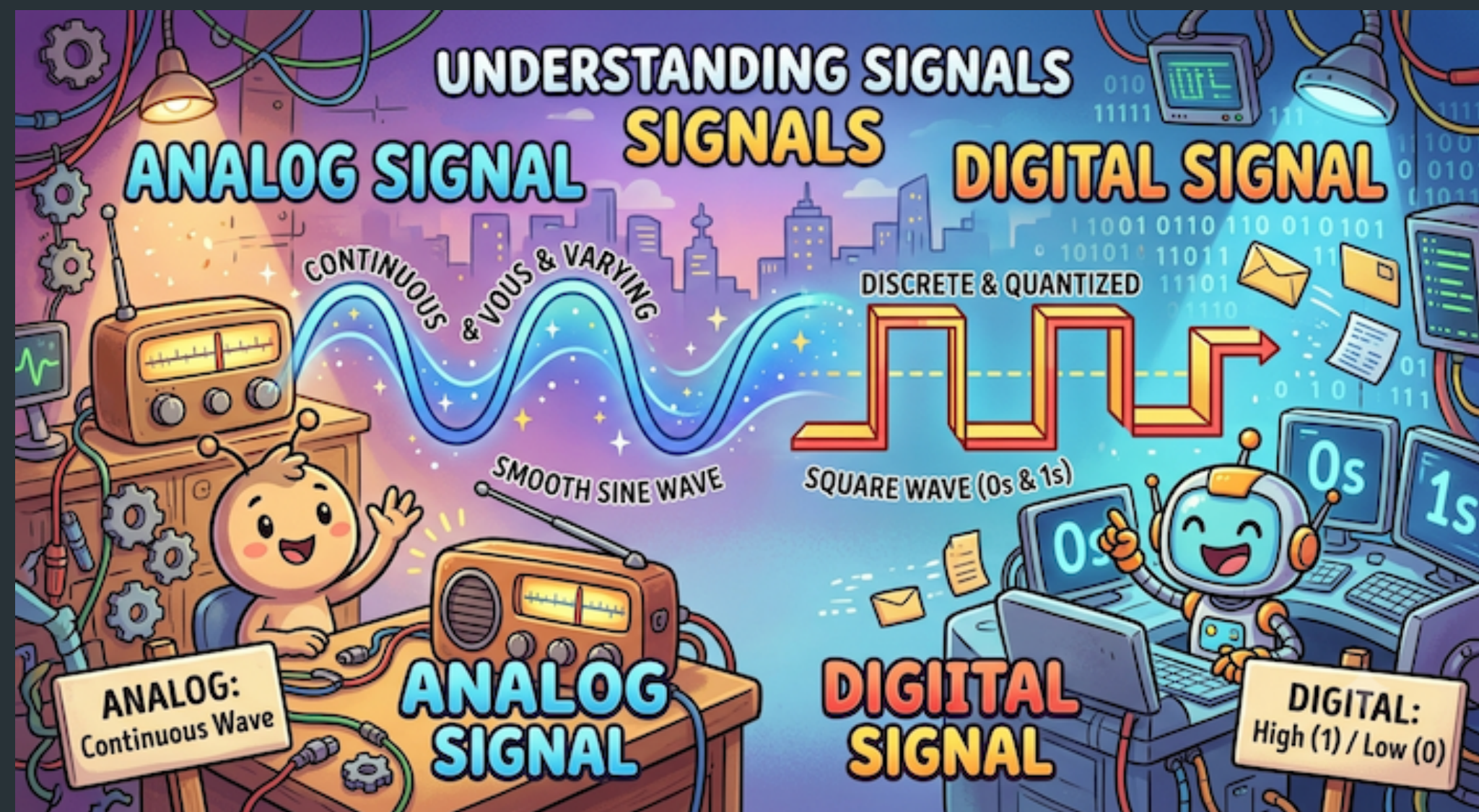


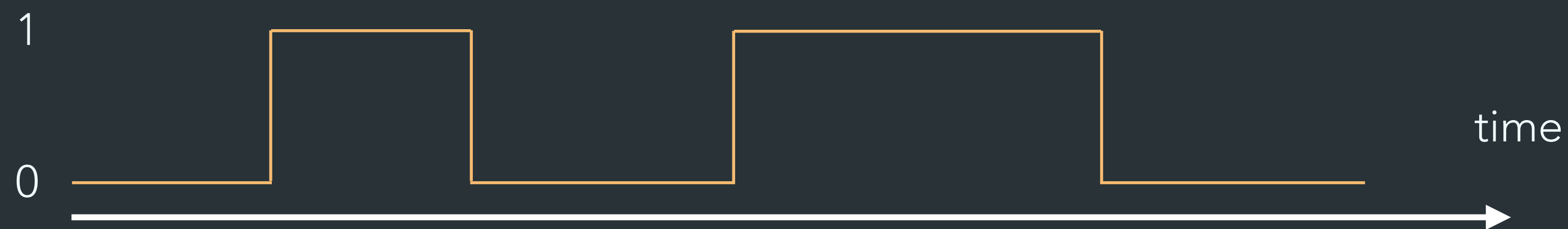
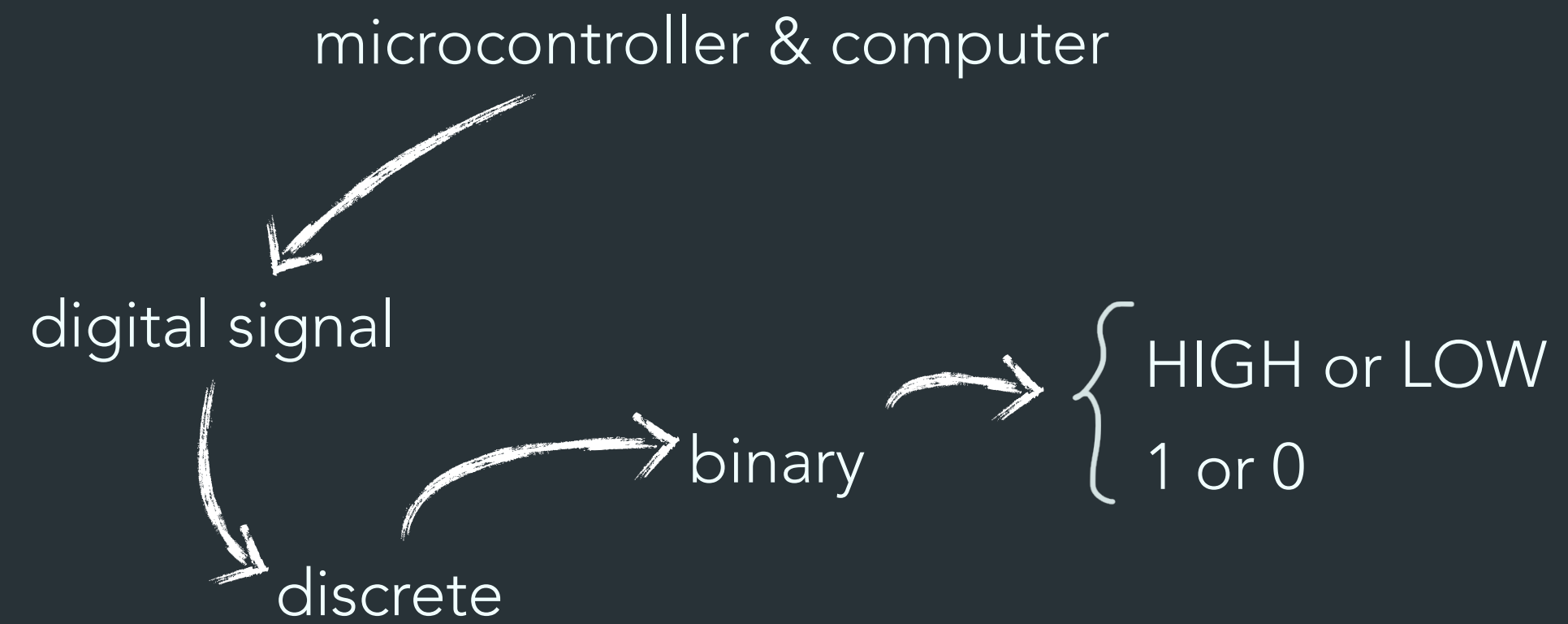


generated with Nano Banana 2

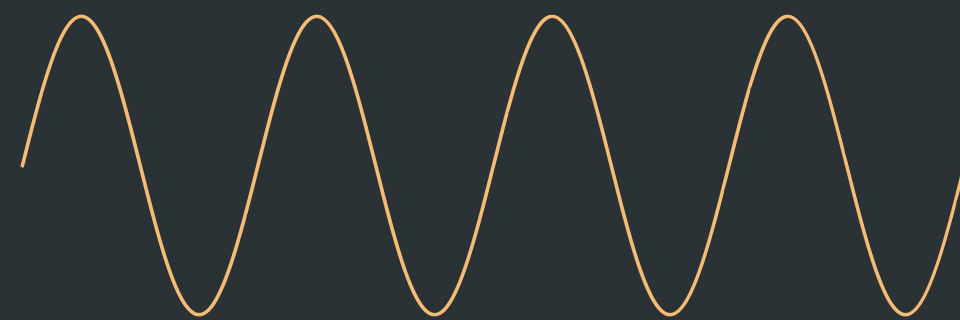
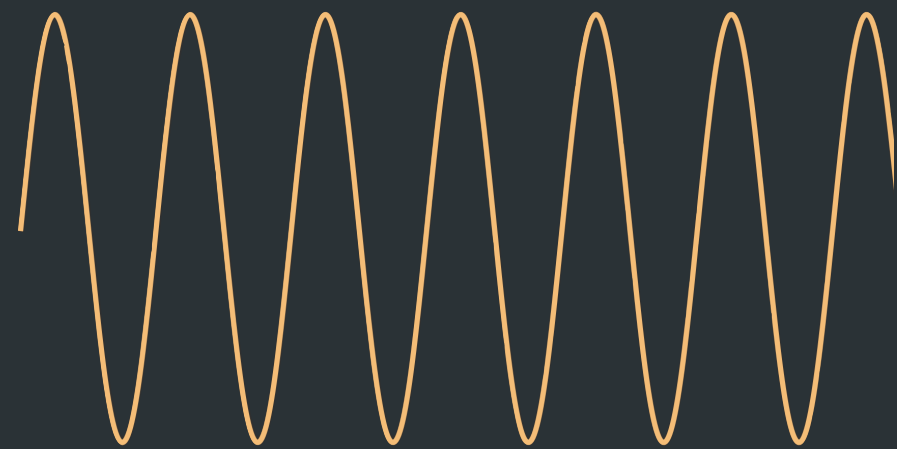
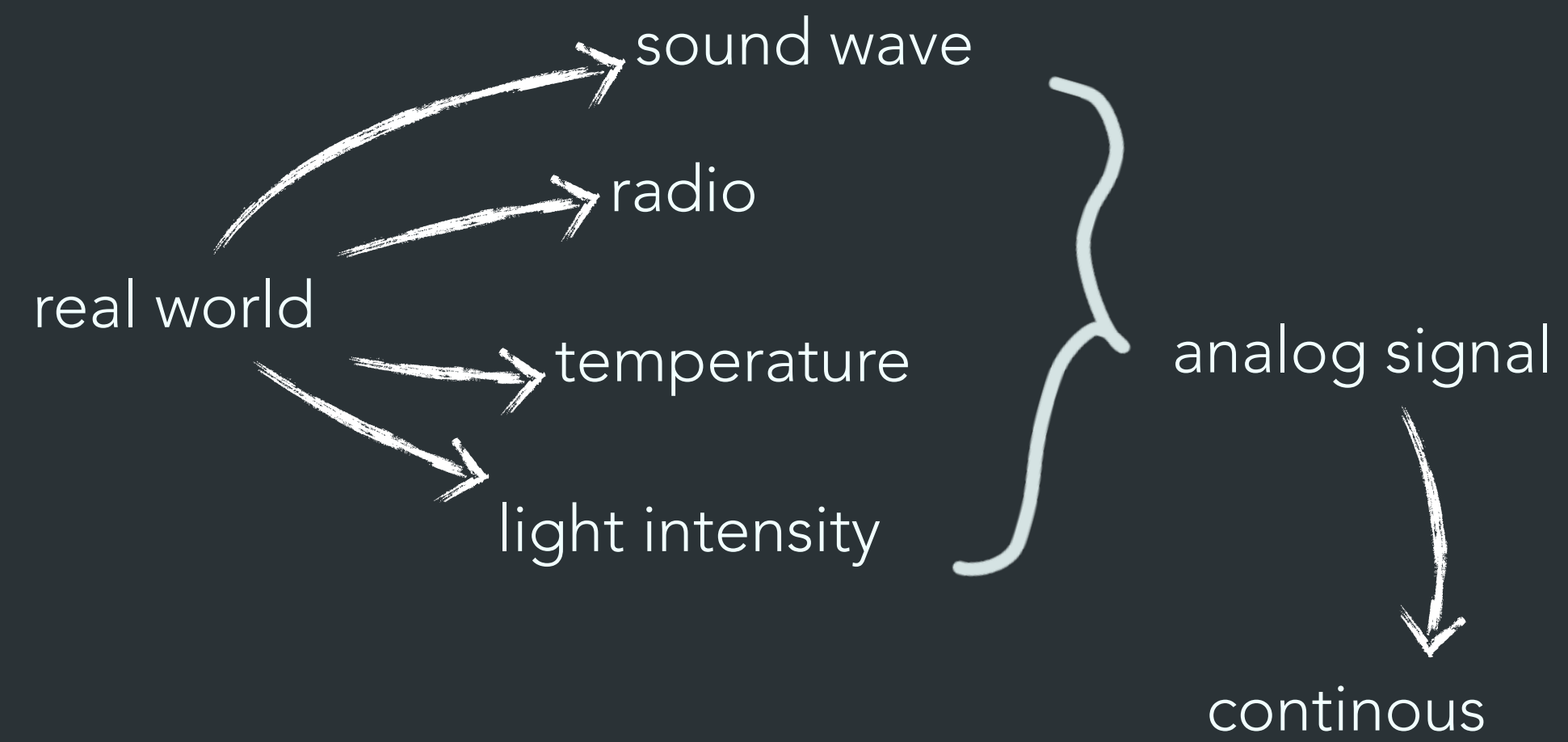
kokchun giang

edge computing -
analog and digital
signals

what is **digital signal**?



what is **analog signal**?



use **ADC** to convert analog signals to digital for microcontroller to process



temperature
sensor



ex

* TMP37

* Picos internal
temp sensor

voltage
(analog)

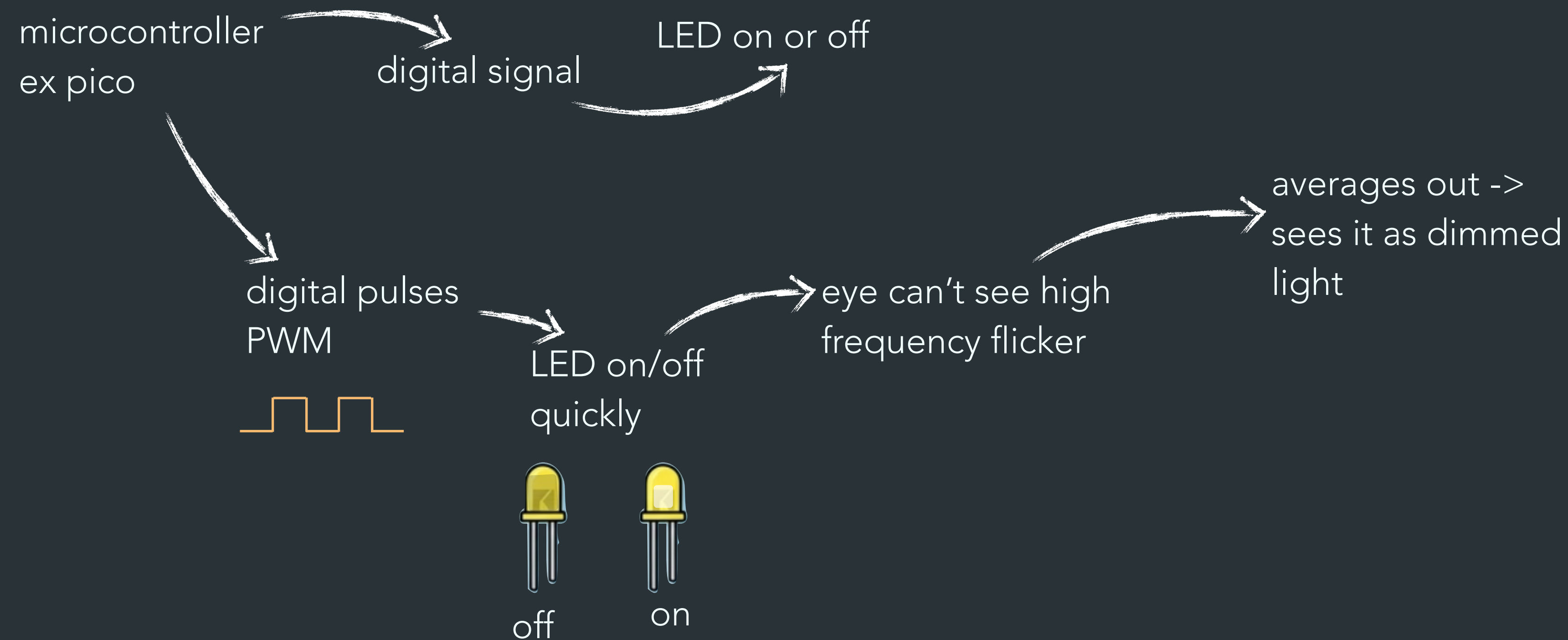
Pico ADC

16 bit
representation
0-65535

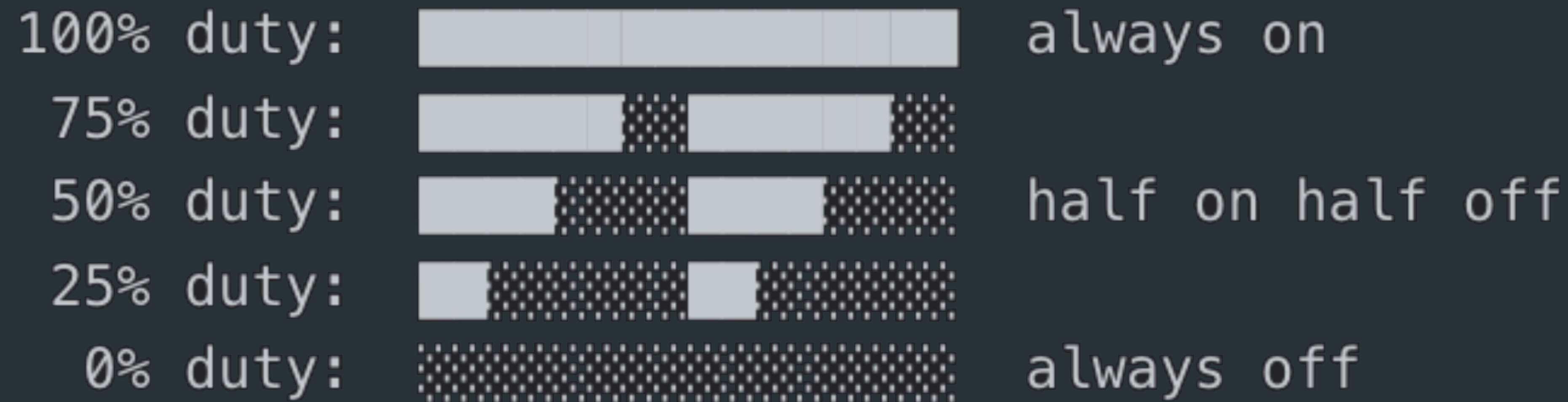
post process to
convert to °C

digital sensors ex
DHT11 and DS18B20
have built in ADC

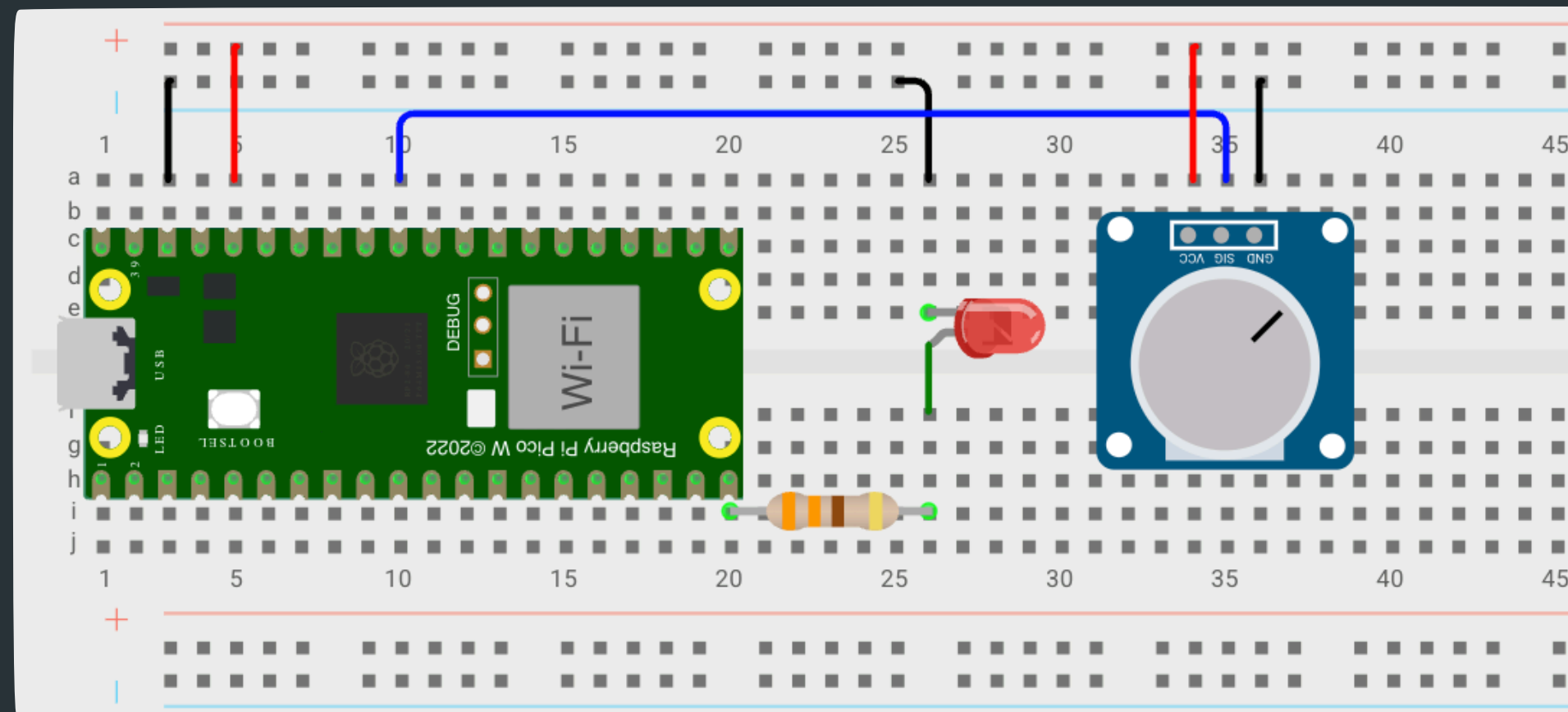
simulate analog signal with pulse width modulation **PWM**



signals **duty cycle** is the percentage of time
it's on in PWM



creating a dimmer with **potentiometer**



potentiometer
acts as **voltage**
divider

